BCSE302P – Database Systems Lab

Project Rubrics

Review 1 (10 Marks)

Problem Definition, Requirement Analysis & Database Design

Objective: Evaluate the understanding of the problem and the suitability of the chosen NoSQL database.

Criterion	Marks
Problem statement, objectives, and scope	2
Selection and justification of NoSQL database (MongoDB/Neo4j/Cassandra, etc.)	3
Data model design (Collections/Graph Schema, ER-to-NoSQL mapping, node/relationship design)	5
Total	**10**

Expected Deliverables

- Problem statement
- Requirement specification
- Database schema (collections or graph model)
- Sample dataset design

Review 2 (10 Marks)

Database Implementation & Core Functionalities

Objective: Assess implementation progress and effective use of NoSQL features.

Criterion	Marks
Database implementation (collections/nodes, relationships, sample data)	2
CRUD operations completed	3
Advanced NoSQL features (aggregation, graph traversal, indexing, transactions, etc.)	5

Criterion	Marks
Total	**10**

Expected Deliverables

- Database created
- Sample data loaded
- CRUD operations working
- Minimum 50 sample records (or 50+ graph nodes)
- Intermediate demonstration
- Source code repository

Review 3 (10 Marks)

Final Demonstration, Evaluation & Documentation

Objective: Evaluate the completed system and presentation.

Criterion	Marks
Complete working demonstration	3
Advanced queries/analytics (aggregation, recommendation, shortest path, etc.)	2
User interface and usability	3
Presentation, viva voce, and response to questions	2
Total	**10**

Expected Deliverables

- Fully functional application
- Final report
- Source code
- User manual
- Presentation slides

- Demonstration video (optional)

Suggested Deliverables Based on Database Type

MongoDB Projects	**Neo4j Projects**
Collections and document schema	Graph schema (nodes, relationships)
CRUD operations	Node and relationship CRUD
Aggregation pipelines	Cypher queries
Indexes	Graph indexes and constraints
Document validation	Graph constraints
Reports and dashboards	Graph visualizations

Minimum Technical Requirements

MongoDB-Based Projects

- At least **5 collections**
- **100+ sample documents**
- CRUD operations
- **5 aggregation queries**
- **2 indexes**
- Basic authentication
- Responsive UI

Neo4j-Based Projects

- At least **5 node labels**
- **6 relationship types**
- **100+ nodes**
- **10 meaningful Cypher queries**
- At least **2 graph algorithms or graph analytics features** (e.g., shortest path, PageRank, community detection, similarity recommendations)

- Interactive graph visualization
- Responsive UI

Overall Evaluation Summary

Review	Focus	Marks
Review 1	Problem definition, requirement analysis, database design	10
Review 2	Database implementation, CRUD, NoSQL features, integration	10
Review 3	Final demonstration, analytics, documentation, presentation	10
Total		**30 Marks**

This rubric ensures that students are assessed not only on the final product but also on **planning, database design, incremental implementation, and effective use of NoSQL-specific capabilities**, making it suitable for both **document databases (MongoDB)** and **graph databases (Neo4j)**.